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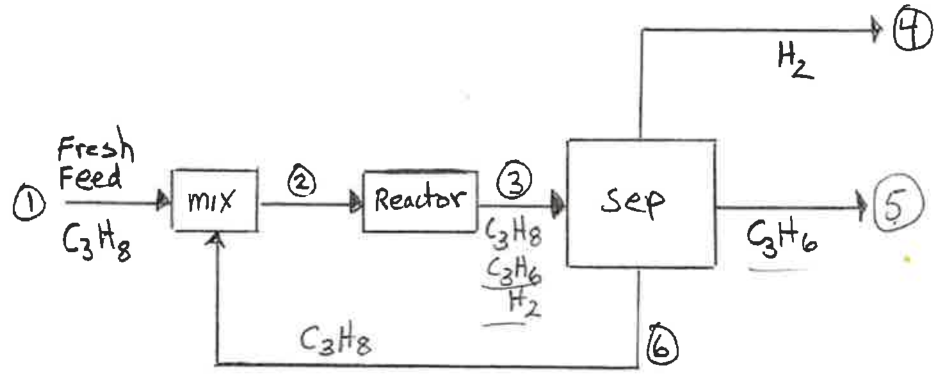
ChE 210 Exam #2 5 November 2025

[1] (30 pts.) A process for propane dehydrogenation is shown in the figure. Assume that the only reaction taking place in the reactor is: $C_3H_8 \rightarrow C_3H_6 + H_2$ and there are no side reactions. There are six streams and the chemical species in each stream are labeled. The single-pass conversion for the reactor is 20% based on propane. The product flow rate is $\dot{n}_5 = 50$ mol/hr for Stream 5. Calculate all of the species molar flow rates and fill in your answers at the bottom of the page.



$n_{2,P} = n_{1,P} + n_{6,P}$

$n_{5,C_3H_6} = 50 = n_{3,C_3H_6}$

RX:

$n_{3,P} = n_{2,P} - \xi_1$

$\xi_1 = 50$

$n_{3,C_3H_6} = \xi_1$

$f_{CO}^{SP} = .2 = \dfrac{\xi_1}{n_{2,P}} \rightarrow n_{2,P} = \dfrac{\xi_1}{.2} = \dfrac{50}{.2} = 250$

$n_{3,H} = \xi_1 = 50$

$n_{3,P} = 250 - 50 = 200$

$n_{6,P} = n_{3,P}$

$n_{1,P} = n_{2,P} - n_{6,P} = 250 - 200 = 50$

$\dot{n}_1 =$ 50 mol/hr, $\dot{n}_2 =$ 250 mol/hr, $\dot{n}_{3,C3H8} =$ 200 mol C_3H_8/hr,

$\dot{n}_{3,C3H6} =$ 50 mol C_3H_6/hr, $\dot{n}_{3,H2} =$ 50 molH_2/hr, $\dot{n}_4 =$ 50 mol/hr,

$\dot{n}_6 =$ 200 mol/hr

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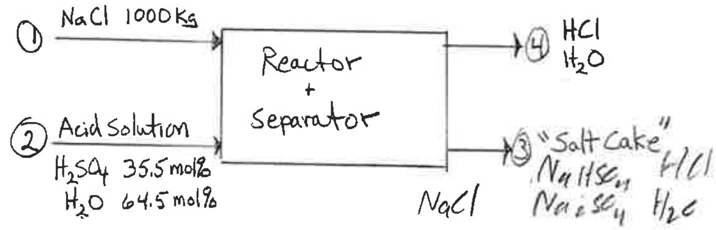
[2] (50 pts.) Salt (NaCl) and sulfuric acid (H_2SO_4) are reacted to produce hydrochloric acid (HCl) in a process represented by two chemical reactions:

$$NaCl + H_2SO_4 \rightarrow NaHSO_4 + HCl \quad \text{(R1)}$$
$$NaCl + NaHSO_4 \rightarrow Na_2SO_4 + HCl \quad \text{(R2)}$$



For the process, there are two input streams: **Stream 1**: 1000 kg of pure NaCl and **Stream 2**: a 35.5 mol% aqueous solution of sulfuric acid. There are two output streams: **Stream 3**: a "Salt Cake" consisting of 80.1 mol% of Na_2SO_4, 5.0 mol% of $NaHSO_4$, 4.2 mol% of NaCl, 9.3 mol% of H_2O, and 1.4 mol% of HCl; and **Stream 4**: a vapor stream of HCl and H_2O.

Species	Molar Mass
Na_2SO_4	142.1
$NaHSO_4$	120.1
NaCl	58.5
H_2O	18.0
HCl	36.5
H_2SO_4	98.1

(a) Calculate the mole of NaCl entering the process in kmol. (5 pts.)

$1000 \text{ kg NaCl} \cdot \frac{\text{kmol}}{58.1 \text{ kg}} = 17.21 \text{ kmol NaCl}$

$n_1 =$ 17.21 kmol ✓ 5

(b) Write species mole balance equations and/or atomic balance equations. The problem can be solved by either method. (20 pts.)

out = in − $\nu\xi$ $\qquad (10+2)-(7+5+0+6)=0$

NaCl: $n_{3,NaCl} = 17.21 - \xi_1 - \xi_2$ ✓

H_2SO_4: $0 = n_{2,H_2SO_4} - \xi_1 \rightarrow n_{2,H_2SO_4} = 15.4 \quad n_2 =$ ✓

H_2O: $\underline{n_{3,H_2O}} + n_{4,H_2O} = n_{2,H_2O}$ ✓

Na_2SO_4: $n_{3,Na_2SO_4} = \cancel{\xi_1} - \quad \xi_1 = .801 n_3 \rightarrow \xi_1 = 15.43$

$NaHSO_4$: $n_{3,NaHSO_4} = \xi_2 \rightarrow \quad \xi_2 = .05 \cdot n_3 \rightarrow \xi_2 = 0.9635$ ✓

HCl: $n_{4,HCl} - n_{3,HCl} = \xi_1 + \xi_2 \Rightarrow n_{4,HCl} = (.801 \cdot 19.27) + (.014 \cdot 19.27)$

$.042 \cdot n_3 = 17.21 - \xi_1 - \xi_1$

$0.042 n_3 + 0.801 n_3 + .05 n_3 = 17.21$?

$n_3 = \cancel{19.27}$ ✓

1	30
2	37
3	20
T	87

20

PROBLEM [2] CONTINUED

(c) Calculate the molar amounts of Streams 2, 3, and 4; and the mole fraction of HCl in Stream 4. (15 pts.)

← calcs on Part b.

$0 = n_{2,H_2SO_4} - \xi_1$

$\xi_1 = .335 n_2$

10

$n_2 = \frac{15.435}{.355} = 43.47$

$n_{4,HCl} = \xi_2 + \xi_1 + n_{3,HCl} = 15.435 + (.014 \cdot 14.27) + (0.05 \cdot 14.27)$
$= 16.67$

$n_{4,H_2O} = (.645 \cdot 43.47) - (14.27 \cdot .093)$
$= 26.24$

$n_4 = 15.7 + 26.24$

$n_2 =$ ~~43.47~~ kmol, $n_3 =$ ~~14.27~~ kmol, $n_4 =$ ~~42.4~~ kmol, $x_{4,HCl} =$ ~~0.384~~

(d) Check the mass balance for the process. That is, show that mass in equals mass out. (10 pts.)

$NaCl$: $14.27 \cdot .642 = 17.21 - 1544 - 0.9635 \rightarrow 0$ ✓

2

H_2SO_4: $0 = (.355 \cdot 43.47) - 15.44 = 0$ ✓

H_2O: $0 = (46.07 \cdot .355) - (14.27 \cdot .093) - (27.9) \Rightarrow 0$ ✓

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$NaHSO_4$: $0 = n_{3\,NaHSO} - \xi_2 = (.05 \cdot 14.27) - 0.9635 = $ ✓

HCl: $0 = \xi_1 + \xi_2 + n_{3HCl} - n_{4,HCl} = 15.43 + 0.9635 + (0.014 \cdot 14.27) - 16.67 = 0$ ✓

→ all mass balances equal zero

Na_2SO_4: $0 = 15.43 - .861 \cdot 14.27 = 0$ ✓

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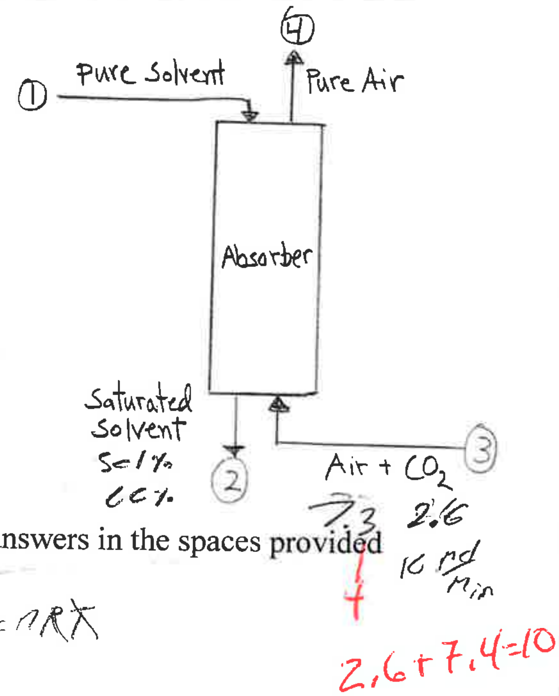
[3] (20 pts.) Carbon dioxide (CO_2) can be removed from air using an absorption process. In the figure, **Stream 1** is a pure solvent stream and **Stream 3** is an air/CO_2 mixture containing 26 mol% CO_2 at a flow rate of 10 kmol/min. The operating temperature and pressure is 50°C and 1 atm. Assume Stream 2 and Stream 3 are in equilibrium at 50°C.



Equilibrium Data at 50°C:

Partial Pressure of CO_2 (mmHg) in Air/CO_2	Liquid Concentration (moles of CO_2/mole of solvent)
668.4	0.680
242.3	0.562
183.8	0.548
71.0	0.489
10.2	0.302

Calculate the molar flow rates and compositions of all streams. Write your answers in the spaces provided below.

bal.

Sol: $n_{2,S} = n_{1,S}$

Air: $n_{4,A} = 7.3$

CO_2: $n_{2,C} = 2.6$

$P_i = yP$

$= .26 \cdot 760$

$= 197.6$

$PV = nRT$

$P = n$

$\frac{mol\, CO_2}{mol\, sol} = 0.550$

$n_{2,C} = 2.6$

$n_{2,S} = n_1$

$n_1 = \frac{2.6}{0.55} = 4.7$

$\frac{2.6}{n_{2,S}} = .550$

$n_1 = n_{2,S} = 4.7$

$n_2 = 4.7 + 2.6 = 7.3$

$x_{CO_2} = \frac{2.6}{7.3} = 0.35$

$\dot{n}_1 =$ 4.7 molS/min, $\dot{n}_2 =$ 7.3 mol/min, $\dot{n}_4 =$ 7.3 (7.4) molAir/min,

$x_{CO2} =$ 0.355 molCO2/mol

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